SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Constituent of Symbiosis International (Deemed University), Pune

Detailed ESE Project Rubrics.

Course Details

Department:	AIML Department	**Batch:**	2024-28
Course Name:	Applications and Use Cases of Machine Learning	**Course Code:**	TE7483
Year / Sem:	Semester 5	**Credit:**	2
Faculty Members:	Dr. Aniket Shahade, Dr. Shrikishna Kolhar, Dr. Sumanto Dutta		

Performance Level Conversion

Performance Level	Description	Awarded Portion
Excellent	Complete, correct, technically sound, well-justified, and clearly presented response.	100%
Good	Mostly correct response with minor gaps in explanation, depth, formatting, or justification.	75%
Satisfactory	Partially correct response showing basic understanding but with noticeable missing details.	50%
Poor	Weak or incomplete response with major conceptual, technical, or presentation gaps.	25%
Not Acceptable	Incorrect, irrelevant, copied, blank, not submitted, or not aligned with the assessment requirement.	0%

ESE: AUCML Case Study Project

Assessment Type: Final Case Study Project Evaluation **Max Marks:** 30 **Weight:** 60%

CO Mapping: CO1, CO2, CO3, CO4

The ESE evaluates the complete development and presentation of the Machine Learning case-study project. Students are expected to demonstrate problem understanding, dataset handling, methodology, implementation or analytical execution, evaluation, interpretation of results, final report quality, and individual understanding through presentation and viva/project defense.

ESE Component-wise Distribution

Component	Marks
Problem Refinement and Use Case Relevance	3
Dataset Understanding, Preprocessing, and Feature Handling	4
Methodology and ML Pipeline Design	5
Implementation / Project Execution	5
Results, Evaluation Metrics, and Comparative Analysis	5
Final Report Quality and Documentation	3
Communication Skills and Presentation	2
Viva / Project Defense	3
Total	**30**

Detailed Rubric for ESE

Component	Marks	Criteria
Problem Refinement and Use Case Relevance	3	**Excellent:** Problem is clearly refined from Phase-I; objective, scope, application domain, users/stakeholders and real-world relevance are strongly explained. **Good:** Problem and relevance are clear, but scope or stakeholder/application discussion is limited. **Satisfactory:** Problem is understandable but lacks refinement, depth or clear connection to a real-world use case. **Poor:** Problem is vague, broad or weakly connected to Machine Learning applications. **Not Acceptable:** Problem is missing, irrelevant or unrelated to the subject.
Dataset Understanding, Preprocessing, and Feature Handling	4	**Excellent:** Dataset is clearly described with source, size, attributes/features, target variable, data types, missing values, preprocessing steps, feature selection/engineering and validation strategy. **Good:** Dataset and preprocessing are mostly clear, but some details such as feature handling, missing values or validation strategy are incomplete. **Satisfactory:** Dataset is described generally; preprocessing is basic and lacks explanation. **Poor:** Dataset usage is unclear; preprocessing is weak, incorrect or not aligned with the ML task. **Not Acceptable:** No valid dataset, data source or data handling is presented.
Methodology and ML Pipeline Design	5	**Excellent:** Provides a complete ML pipeline including data preparation, model selection, training strategy, evaluation method and justification of chosen algorithms. **Good:** Pipeline is mostly complete but lacks detailed justification or misses one important stage. **Satisfactory:** Basic methodology is present but model choice, evaluation plan or workflow is weakly justified. **Poor:** Methodology is incomplete, technically weak or not suitable for the problem. **Not Acceptable:** No meaningful methodology is provided.
Implementation / Project Execution	5	**Excellent:** Implementation/prototype is complete and aligned with the proposed methodology. Code, workflow, model execution, outputs and project structure are clear and reproducible. **Good:** Implementation is mostly complete but has minor gaps in structure, reproducibility or explanation. **Satisfactory:** Partial implementation is shown; output is available but execution details are limited. **Poor:** Implementation is incomplete, poorly organized or only partially functional. **Not Acceptable:** No implementation, prototype, analytical execution or working evidence is provided.
Results, Evaluation Metrics, and Comparative Analysis	5	**Excellent:** Results are accurate, clearly presented and analyzed using suitable metrics such as accuracy, precision, recall, F1-score, confusion matrix, RMSE, MAE, ROC-AUC, clustering metrics or task-appropriate measures. **Good:** Results and metrics are mostly correct but analysis or comparison is limited. **Satisfactory:** Results are present but weakly analyzed or only partially connected to the objective. **Poor:** Results are incomplete, poorly presented or metrics are inappropriate for the ML task. **Not Acceptable:** No result, metric, observation or analysis is provided.
Final Report Quality and Documentation	3	**Excellent:** Report is professional, well-structured, technically clear, complete and includes problem statement, methodology, dataset details, results, conclusion, future scope and references. **Good:** Report is mostly complete but contains minor formatting, language or documentation issues. **Satisfactory:** Report contains essential sections but lacks depth, clarity or proper formatting. **Poor:** Report is incomplete, poorly organized or lacks important sections. **Not Acceptable:** Report is missing, copied, irrelevant or not submitted in required format.

Component	Marks	Criteria
Communication Skills and Presentation	2	**Excellent:** Presentation is clear, confident, structured, time-managed and effectively uses visuals to explain the project. **Good:** Presentation is mostly clear but has minor issues in flow, confidence or visual explanation. **Satisfactory:** Presentation explains the basic idea but lacks clarity, structure or confidence. **Poor:** Presentation is unclear, unstructured or difficult to follow. **Not Acceptable:** Student is unable to present the work meaningfully.
Viva / Project Defense	3	**Excellent:** Demonstrates deep individual understanding of the project, dataset, model, methodology, results, limitations and future scope; answers all questions correctly and confidently. **Good:** Demonstrates good understanding and answers most questions correctly. **Satisfactory:** Shows basic understanding but struggles with technical depth or follow-up questions. **Poor:** Shows limited understanding and answers only simple questions. **Not Acceptable:** Unable to explain the project or gives irrelevant answers.
Total	**30**	